Sounak Saha

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EDUCATION

B.Tech in Computer Science, Sai University — Sep '21 — Jun '25, Chennai, India

INTERNSHIPS

ML Research Intern — Jul '24 — Aug '24
Jawaharlal Nehru University (JNU) — Delhi, India

- Built and trained a CNN-based facial emotion recognition model using OpenCV preprocessing, implementing data augmentation and evaluation metrics to achieve 0.97+ accuracy.

ML Intern — Jan '25 — Sep '25
ERIC Robotics — Pune, India

- Prepared training datasets and trained a YOLOv11 object detection model, evaluating performance using mAP and optimizing inference speed for robotic environments.
- Conducted extensive testing and validation of the model using Flask for deployment on robotic platforms.
- Optimized the detection pipeline to enhance computational efficiency for deployment in constrained robotic environments.

AI Intern — Sep '25 — Dec '25
ThinkZone — Vizag, India (Remote)

- Built and deployed image-processing pipelines using **Gemini 2.5 Pro**, enabling automated extraction, formatting, and structuring of data from complex document images.
- Developed **Flask-based backend APIs** to integrate AI image-processing results with the front-end application.
- Designed and implemented **MongoDB logging workflows** to store inference outputs, request metadata, and system diagnostics for debugging and auditability.

PROJECTS

Multimodal AI Therapy Companion Link

- Built an **asynchronous FastAPI backend** to handle concurrent **STT and TTS pipelines**, reducing perceived response latency for real-time voice interaction.
- Designed an end-to-end **voice-first AI pipeline** integrating **OpenAI APIs** for conversational reasoning and **Edge TTS** for high-quality neural speech synthesis.
- Implemented **personality-driven system prompts** enabling distinct AI personas (*Jo* and *Dave*) with controlled pacing, tone, and emotional response rules.
- Applied **signal-level speech heuristics** (pauses, pacing, turn-taking) to improve conversational naturalness and user comfort.
- Deployed the Dockerized application to Hugging Face Spaces, configuring environment variables, dependency management, and runtime constraints for stable public inference.

Agentic AI-Research Assistant Link

- Designed and owned end-to-end architecture of a full-stack AI research assistant, including backend services, vector retrieval layer, frontend interface, and containerized deployment.
- Designed and implemented a **RAG pipeline** using **Pinecone vector database**, enabling retrieval of context-aware answers from uploaded PDFs and historical queries.
- Engineered **document ingestion and summarization** with OCR (Tesseract) and PDF parsing, creating chunked embeddings for accurate follow-up responses.
- Designed multi-step conditional workflows with tool-calling and state management.
- Integrated external APIs and structured JSON pipelines for task automation.
- **Containerized the system using Docker**, ensuring reproducible environments for backend, frontend, and LLMs, simplifying deployment on any machine.
- Handled async event-driven logic for real-time task execution.

SKILLS

Languages: Python, JavaScript, SQL
Machine Learning: Scikit-learn, Feature Engineering, Model Evaluation, Supervised Learning, Unsupervised Learning
Deep Learning: PyTorch, TensorFlow, Computer Vision, NLP
Data & Databases: MongoDB, Pinecone, PostgreSQL, Pandas
Backend / APIs: Flask, FastAPI, REST APIs, Async Processing
Cloud: Docker, Git, Github Actions, Hugging Face Spaces